Online Learning Management System with Performance Analytics for Senior High School Students

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In partial fulfillment of the requirements for the degree of

Bachelor of Science in Information Technology

by:

Rivera, Joshua K.

Lim, Patricia G.

Ocampo, Daniel S.

Rebecca R. Fajardo, MSEE

Adviser

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ABSTRACT

Title: Online Learning Management System with Performance Analytics for Senior High School Students

Researchers:

Rivera, Joshua K.

Lim, Patricia G.

Ocampo, Daniel S.

Degree: Bachelor of Science in Information Technology

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Project Description:

This study presents the development of a web-based Online Learning Management System (LMS) integrated with performance analytics to enhance the teaching and learning experience for senior high school students. Traditional classroom methods and basic online platforms often lack centralized management, real-time performance tracking, and detailed insights into student progress.

The proposed system allows teachers to upload learning materials, create assessments, and monitor student performance, while students can access lessons, submit assignments, and track their academic progress. The system includes analytics features that evaluate student performance trends, identify learning gaps, and generate reports to support academic decision-making.

The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Results indicate that the system improves student engagement, supports data-driven instruction, and enhances overall learning outcomes.

CHAPTER 1: INTRODUCTION

I. Introduction

Education has evolved significantly with the integration of digital technologies, especially in response to the growing demand for flexible and accessible learning environments. However, many schools still face challenges in managing online classes, tracking student performance, and maintaining organized learning resources.

Learning Management Systems (LMS) provide a centralized platform for delivering educational content, managing assessments, and monitoring student progress. When combined with performance analytics, these systems can offer valuable insights into student learning behavior and academic performance.

This study proposes an Online Learning Management System with performance analytics to improve teaching efficiency, enhance student engagement, and support data-driven educational practices.

II. Significance of the Study

This system will benefit the following:

- Students – Easy access to learning materials and performance tracking
- Teachers – Efficient class management and assessment tools
- School Administrators – Centralized monitoring of academic performance
- Educational Institutions – Improved quality of education through data insights

III. Statement of the Problem

General Problem

How can an LMS with performance analytics improve teaching and learning outcomes for senior high school students?

Specific Problems

1. How to provide a centralized platform for learning materials and assessments?
2. How to monitor and evaluate student performance effectively?
3. How to identify learning gaps using performance data?
4. How to generate reports for academic monitoring and decision-making?

IV. Scope and Limitations

Scope

- Online course and content management
- Assignment submission and assessment
- Performance analytics and reporting
- User roles: Admin, Teacher, Student

Limitations

- Requires stable internet access
- Limited to academic use within the institution
- Does not include advanced AI-based tutoring features

OPERATIONAL DEFINITION OF TERMS

Administrator – The personnel responsible for managing system users and configurations.

Assessment – Activities such as quizzes, exams, and assignments used to evaluate student learning.

Course Management – The process of organizing and delivering learning materials and activities.

Database – A structured system that stores user data, course content, and performance records.

Frontend – The interface where users interact with the system.

Backend – The server-side component responsible for processing data and system operations.

Learning Management System (LMS) – A platform used to deliver, manage, and track educational content and performance.

Performance Analytics – The analysis of student data to evaluate academic progress and trends.

Submission System – A feature that allows students to submit assignments online.

Report Generation – The creation of summaries and insights based on student performance data.

Role-Based Access Control (RBAC) – A system that restricts access based on user roles.

Usability (ISO 25010) – The ease with which users can effectively use the system.

Reliability (ISO 25010) – The system's ability to perform consistently under specified conditions.

Security (ISO 25010) – Measures to protect academic and user data.